



MILAN KUMAR MAHARANA

SENIOR LEAD ENGINEER

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PROFESSIONAL OVERVIEW

Senior Lead Engineer with 8+ years of experience designing and delivering scalable, enterprise-grade applications using **Java, Spring Cloud, Spring Boot, and Spring AI** across BFSI, eCommerce, CRM, and other business domains. Skilled in building cloud-native microservices on **AWS**, developing AI-powered solutions with **LLM integrations**, and creating secure, high-performance APIs that support large-scale business operations. Proven track record of delivering production-ready solutions through **CI/CD pipelines**, Agile development practices, and cross-functional collaboration. Recognized for mentoring teams, driving technical excellence, and helping organizations adopt modern cloud and AI-driven architectures to accelerate innovation and business value.

CORE TECHNICAL SKILLS

Languages & Frameworks: Java (Core, Collections, Streams) | Spring AI, Spring Cloud, Spring Boot, Spring Framework, Spring Data JPA, Hibernate, Spring Security, LLM clients, RAG (Retrieval-Augmented Generation)

AI/ML Integration: OpenAI API, Google Vertex AI, Ollama, MCP Server

Spring Ecosystem Expertise: Spring Boot Starters | Auto Configuration | Spring Boot Actuator (/health, /metrics, /info) | Spring Cloud Config | Spring Boot DevTools

Microservices & Architecture: Microservices Design Patterns | REST API Development (GET, POST, PUT, DELETE, PATCH) | API Status Codes (1xx-5xx) | Service Registry (Eureka) | Client-Side Load Balancing | Service Discovery | Configuration Management (YAML/Properties) | Multi-tenancy Implementation

Inter-Service Communication: RestTemplate | OpenFeign | WebClient (Reactive) | RestClient (Spring6) | HTTP Interfaces | Synchronous & Asynchronous Communication

API Gateway & Routing: Spring Cloud Gateway | Request Routing | Load Balancing | Authentication & Authorization | Rate Limiting | API Aggregation

Databases & ORM: MySQL, Oracle, PostgreSQL, MongoDB, Vector Databases (PGVector, Pinecone, MongoDB Atlas, Chroma) | JDBC, Hibernate, JPA | Schema Design | Transaction Management

Cloud & Containerization: AWS (EC2, S3) | Docker (Images, Containers, Dockerfile, DockerCompose) | Docker Registry

DevOps & Deployment: Maven, Git, Jenkins, GitLab CI/CD | Docker Build

Observability & Monitoring: Logging | Distributed Tracing (Zipkin, Jaeger) | Spring Boot Actuator Endpoints | Health Checks

Asynchronous Processing & Messaging: RabbitMQ (Direct, Fanout, Topic Exchanges) | Apache Kafka (Event Streaming, Topics, Partitions) | Spring Cloud Stream | Message Queue Patterns | Event-Driven Architecture

Security & Authentication: Spring Security | OAuth 2.0 (Authorization Code, Client Credentials) | OIDC (OpenID Connect) | JWT Tokens | Keycloak Integration | Role-Based Access Control (RBAC) | Password Encryption (BCrypt) | OKTA

Resilience & Fault Tolerance: Resilience4j (Circuit Breaker, Retry, Rate Limiter, Bulkhead) | TimeoutManagement | Graceful Degradation | Fallback Strategies | Health Checks & Recovery

Testing & Quality: JUnit 5, Mockito | Integration Testing | Unit Testing | SOLID Principles | Clean Code | Design Patterns

Tools & IDEs: Eclipse, IntelliJ IDEA, Spring Tool Suite (STS) | Log4J | Monitoring Tools

Client-side / UI: Basic React | JavaScript | TypeScript | HTML5 | CSS3 | Responsive Design | JSP | Thymeleaf | REST API integration

WORK EXPERIENCE

IT Analyst

TATA Consultancy Services Limited, Bengaluru | 12 Sep 2022 - Present

- Write clean, efficient, and maintainable Java code using frameworks such as Spring and Hibernate to translate business requirements into technical solutions.
- Design and implement RESTful and SOAP APIs for inter-service and client communication.
- Develop and maintain data persistence layers, write optimized SQL queries, and manage transactions for scalable database operations.
- Ensure code quality through unit and integration testing (JUnit/Mockito/Spring Test), CI pipelines, and thorough peer code reviews.
- Identify and resolve bugs and performance bottlenecks, applying profiling and optimization techniques to improve throughput and latency.
- Implement secure coding practices and controls (authentication, input validation, encryption) to protect against vulnerabilities.
- Manage source code with Git, produce technical documentation (APIs, architecture), and collaborate with cross-functional teams (design, QA, PM).

Senior Software Engineer

HCL Technologies Pvt Ltd, Bengaluru | 07 Jul 2021 – 01 Sep 2022

- Implement features in Java/Spring: controllers, services, repositories, and API versioning (OpenAPI).
- Automated tests: JUnit 5, Mockito, Spring Boot Test.
- CI quality gates: run tests, SonarQube, and linting in GitHub Actions/Jenkins.
- Agile delivery: user stories, sprint planning, standups, demos.
- Git workflow & reviews: feature branches, PRs, mandatory code reviews, merge strategies.

Staff Engineer

LeadIcon Technologies Pvt Ltd, Bengaluru | 01 Nov 2018 – 30 June 2021

- Design, implement, and maintain Java applications across all development phases, delivering scalable and maintainable solutions.
- Conduct software analysis, develop and test features, and perform debugging to ensure robust, production-ready releases.
- Identify, triage, and resolve issues in production and non-production environments using root-cause analysis and corrective measures to minimize downtime.
- Develop and maintain Java application code following coding standards, design patterns, and industry best practices.

EDUCATION

Gandhi Engineering College, Odisha | B. Tech

University: Biju Patnaik University of Technology, Odisha

YOP: April 2017, **CGPA:** 7.32

Newton College of Science and Technology, Odisha | Intermediate

Board: Council of Higher Secondary Education, Odisha

YOP: April 2013, **Marks:** 59.01%

Govt. High School, Odisha | Matriculation

Board: Higher Secondary Education, Odisha

YOP: April 2010, **Marks:** 63.83%

SKILLS

- Communication
- Teamwork
- Critical thinking
- Time management
- Leadership
- Adaptability

ACHIEVEMENTS

Xcelerate Warrior certificate.

TATA Consultancy Service Ltd
2024

Feedback Enabler certificate.

TATA Consultancy Service Ltd
2025

tcsAI Spark certificate.

TATA Consultancy Service Ltd
2025

Service & Commitment Award.

TATA Consultancy Service Ltd
2025

Xcelerate Victor Certificate.

TATA Consultancy Service Ltd
2026

PROJECTS OVERVIEW

Title: Johnson & Johnson Vision (J&J Vision)

Type: Digital, e-commerce and administrative application

Role: Lead Engineer

Team Size: 12

Description: Johnson & Johnson Vision (J&J Vision), a subsidiary of J&J MedTech, is a global leader in eye health, specializing in contact lenses (ACUVUE®) and surgical ophthalmology (cataract, refractive). The company operates in two main divisions: Vision Care and Surgical Vision, providing solutions for myopia, dry eye, and cataracts.

Roles & Responsibilities: Architecting robust, scalable server-side applications using Java, Spring Boot, and Hibernate. • Building responsive, user-friendly interfaces using JavaScript, React, TypeScript, and modern web components. • Designing, developing, and optimizing Oracle DB for high performance. • Creating and consuming RESTful APIs to ensure seamless integration between front-end and back-end services. • Managing end-to-end lifecycle of software products, from requirements gathering and design to testing and deployment. • Participating in daily stand-ups, sprint planning, and retrospectives in an Agile environment. • Working with UX/UI designers, product managers, clinical experts, and data scientists to translate business requirements into technical specifications. • Writing clean, maintainable code and performing peer code reviews to ensure compliance with company standards. • Developing comprehensive test scripts using frameworks like JUnit. • Identifying, analyzing, and resolving software defects and performance bottlenecks in production and non-production environments. • Managing application deployment using Docker, Kubernetes, and CI/CD pipelines in cloud environments.

Title: Deutsche Bank Project

Type: BFSI (Banking, Financial Services and Insurance sector)

Role: Senior Software Engineer

Team Size: 15

Description: Deutsche Bank is the leading German bank with strong European roots and a global network. The bank focuses on its strengths in a Corporate Bank newly created in 2019, a leading Private Bank, a focused investment bank and in asset management.

Roles & Responsibilities: Develop and maintain web applications using Java 17+, Spring Boot, Spring Data JPA, and Hibernate. Design and build interactive user interfaces using Java Script, and HTML5/CSS3. • Design and optimize Oracle PL/SQL databases. Create secure RESTful APIs services for front-end and back-end communication. • Participate in all Agile ceremonies in Jira, and drive CI/CD pipelines using Jenkins, Git/Bitbucket, Maven, and SonarQube. • Implement Test Driven Development (TDD) and Behavior Driven Development (BDD), writing unit (JUnit/Mockito) and functional tests to ensure high code quality. • Provide L3 support for critical production issues, investigating root causes and providing technical solutions.

Title: Orbyo Project

Type: Customer Relationship Management (CRM) tools

Role: Staff Engineer

Team Size: 8

Description: Orbyo is an AI-powered Customer Data Platform (CDP) and marketing automation tool designed to accelerate sales by unifying internal data with external market insights. It helps businesses analyze, predict, and act on customer behavior, specifically targeting industries like automotive with features like lead scoring, geo-targeting, and dynamic content.

Roles & Responsibilities: Developing and enhancing Orbyo backend functionality using Core Java, J2EE, Spring Boot, and Spring MVC. • Creating RESTful APIs to integrate with external systems, mobile apps, and third-party data sources. • Handling database design, creating complex SQL queries, and optimizing database performance using ORM frameworks like Hibernate or JPA. • Implementing Spring Batch for scheduling large data imports, lead processing, or reporting jobs. • Developing user-friendly interfaces using modern java script and css for visualizing CRM data. • Implementing widgets and UI components to display real-time insights for sales/marketing teams.